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Stochastic Optimization: Introduction

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What is it about?



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- **Rigorous modeling** and **solving** of optimization problems with **stochastic uncertainty**
 - Model practical problems
 - Hands on implementation
 - Classical solution methodologies
- **Prerequisites:** Optimisation in Practice 3GS5020, Vincent's catch up, probability theory, convex analysis



- 2hrs teaching + 1hr exercise
- In-course exercise (theoretical and modeling questions, coding)
- Practical works are recommended to use Python (Pyomo) or Julia
- Slides and supporting materials on centralesupelec.edunao.com
- Office hour: Thursday afternoon 13h00 – 15h00, or by appointment with Zhuojun XIE (xie.zhuojun@centralesupelec.fr sa.249 building Bouygues)



- **Two individual mini practical works (60%)**
 - Submit your models, codes, and result analysis (can be in Jupyter Notebook)
 - Subject will be in Edunao
- **Course competition project (40%)**
 - Group project, maximum 4 persons per group
 - Subject will be presented in the third lecture 25/02
 - Submit your models, codes, and results analysis



- Shapiro, A., Dentcheva, D., & Ruszczyński, A. (2021). *Lectures on stochastic programming: modeling and theory*. Society for Industrial and Applied Mathematics. (newest version in Edunao)
- Birge, J. R., & Louveaux, F. (2011). *Introduction to stochastic programming*. Springer Science & Business Media.



Today's outline



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- An illustrative example
- General form of stochastic programs and a recourse model
- Value of information with the routing example
- Exercise: Pierre's planting plan problem

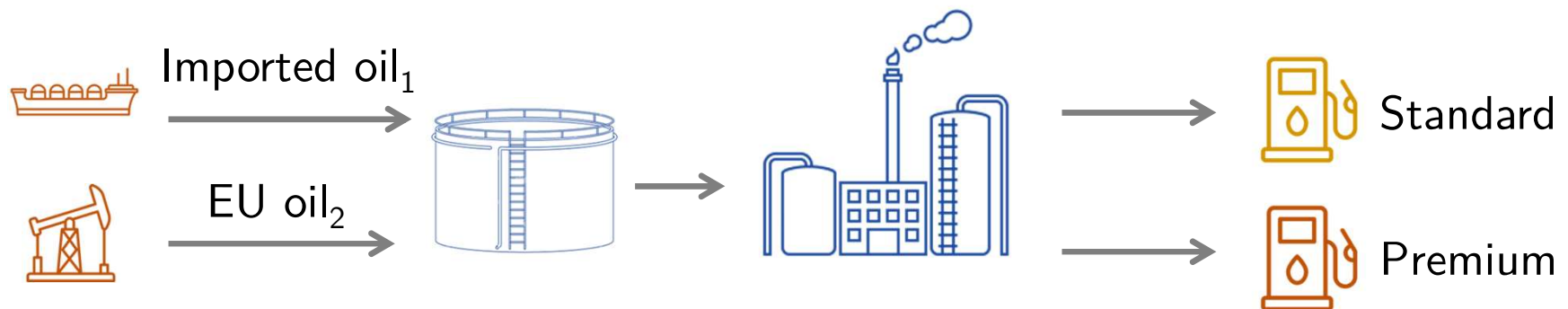


An illustrative example



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- TotalEnergies uses **EU** and **overseas crude oil** to produce **standard** and **premium** gasoline to satisfy the domestic demand



- Production costs: $[c_1, c_2]^\top = [2, 3]^\top$
- Productivity:
 - 1-unit $oil_1 \rightarrow 2$ standard + 3 premium
 - 1-unit $oil_2 \rightarrow 6$ standard + 3 premium
- Demand: $[h_1, h_2]^\top = [180, 162]^\top$
- Production capacity: $oil_1 + oil_2 \leq 100$



An illustrative example



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- The company needs to decide **how much to buy for each oil** to **minimize the production cost while satisfying the demands**
- How to formulate the problem?
- Decision variables: $[x_1, x_2]^\top$

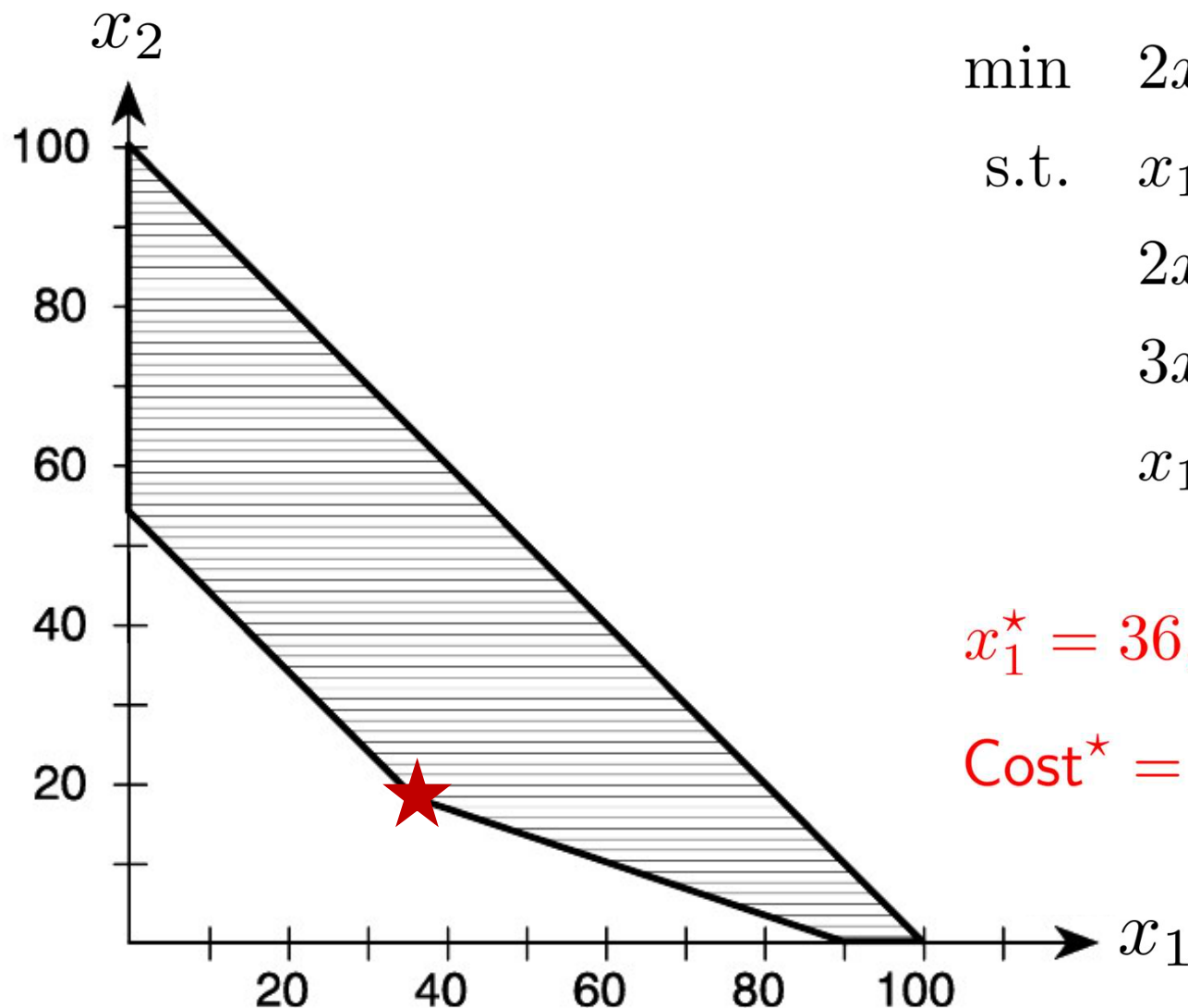
$$\begin{aligned} \min \quad & 2x_1 + 3x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 100 \\ & 2x_1 + 6x_2 \geq 180 \\ & 3x_1 + 3x_2 \geq 162 \\ & x_1, x_2 \geq 0 \end{aligned}$$



An illustrative example



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$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$2x_1 + 6x_2 \geq 180$$

$$3x_1 + 3x_2 \geq 162$$

$$x_1, x_2 \geq 0$$

$$x_1^* = 36, x_2^* = 18$$

$$\text{Cost}^* = 126$$



- **Uncertainty** in
 - **productivity**: gasoline production from the crude oil may change randomly
 - **demand**: weekly demands of the two types of gasoline vary randomly
- Decisions must be **fixed in advance** and cannot be changed during the week
- But uncertainties are only realized and observed afterwards



- Productivity:
 - 1-unit oil₁ → $2 + \eta_1$ standard + 3 premium
 - 1-unit oil₂ → 6 standard + $3.4 - \eta_2$ premium
- Demand:
 - standard: $180 + \theta_1$
 - premium: $162 + \theta_2$
- Assume all random variables are independent and
$$\theta_1 \sim \mathcal{N}(0, 12) \quad \in [-30.91, 30.91] \quad (99\% \text{ CI})$$
$$\theta_2 \sim \mathcal{N}(0, 9) \quad \in [-23.18, 23.18] \quad (99\% \text{ CI})$$
$$\eta_1 \sim \mathcal{U}[-0.8, 0.8]$$
$$\eta_2 \sim \text{Exp}(\lambda = 2.5) \quad \in [0, 1.84] \quad (99\% \text{ CI})$$



- Now, the LP with random variables

$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$(2 + \eta_1)x_1 + 6x_2 \geq 180 + \theta_1$$

$$3x_1 + (3.4 - \eta_2)x_2 \geq 162 + \theta_2$$

$$x_1, x_2 \geq 0$$

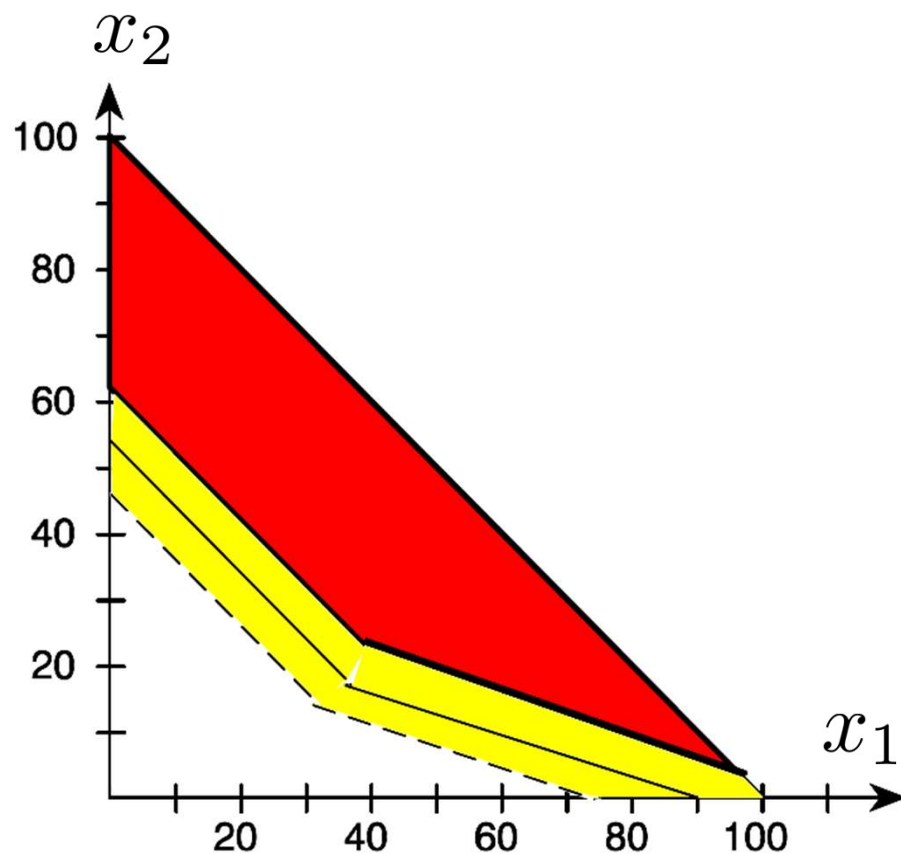
- Is this a well-defined problem?



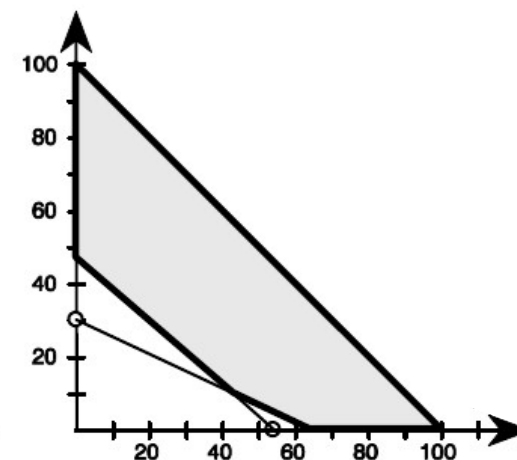
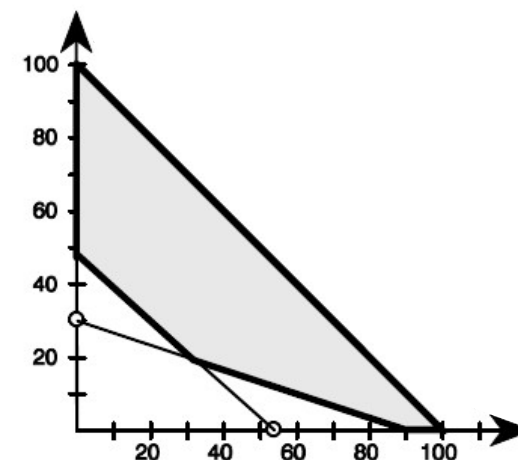
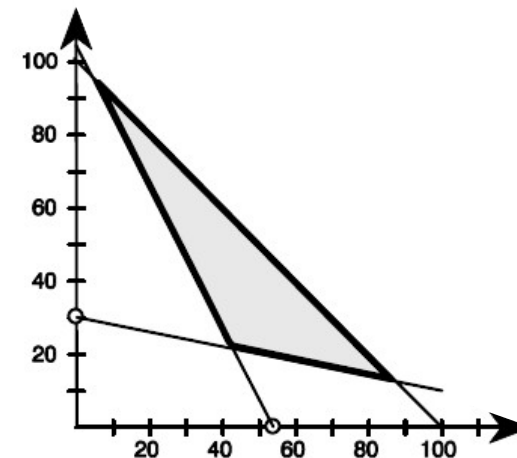
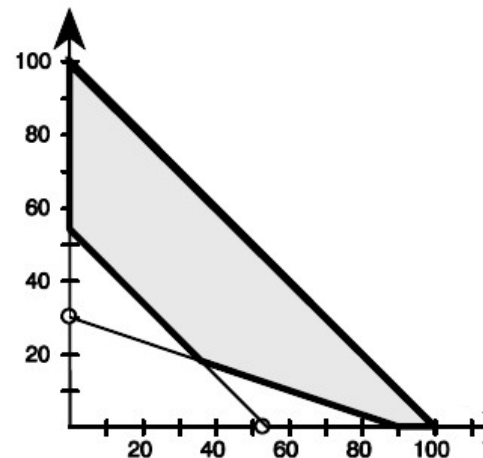
Which one is the feasible set?



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varying with demands θ_1, θ_2



varying with productivities η_1, η_2



- How about forget about the uncertainty, i.e., using the **expected values**?

$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$(2 + \mathbf{0})x_1 + 6x_2 \geq 180 + \mathbf{0}$$

$$3x_1 + (3.4 - \mathbf{0.4})x_2 \geq 162 + \mathbf{0}$$

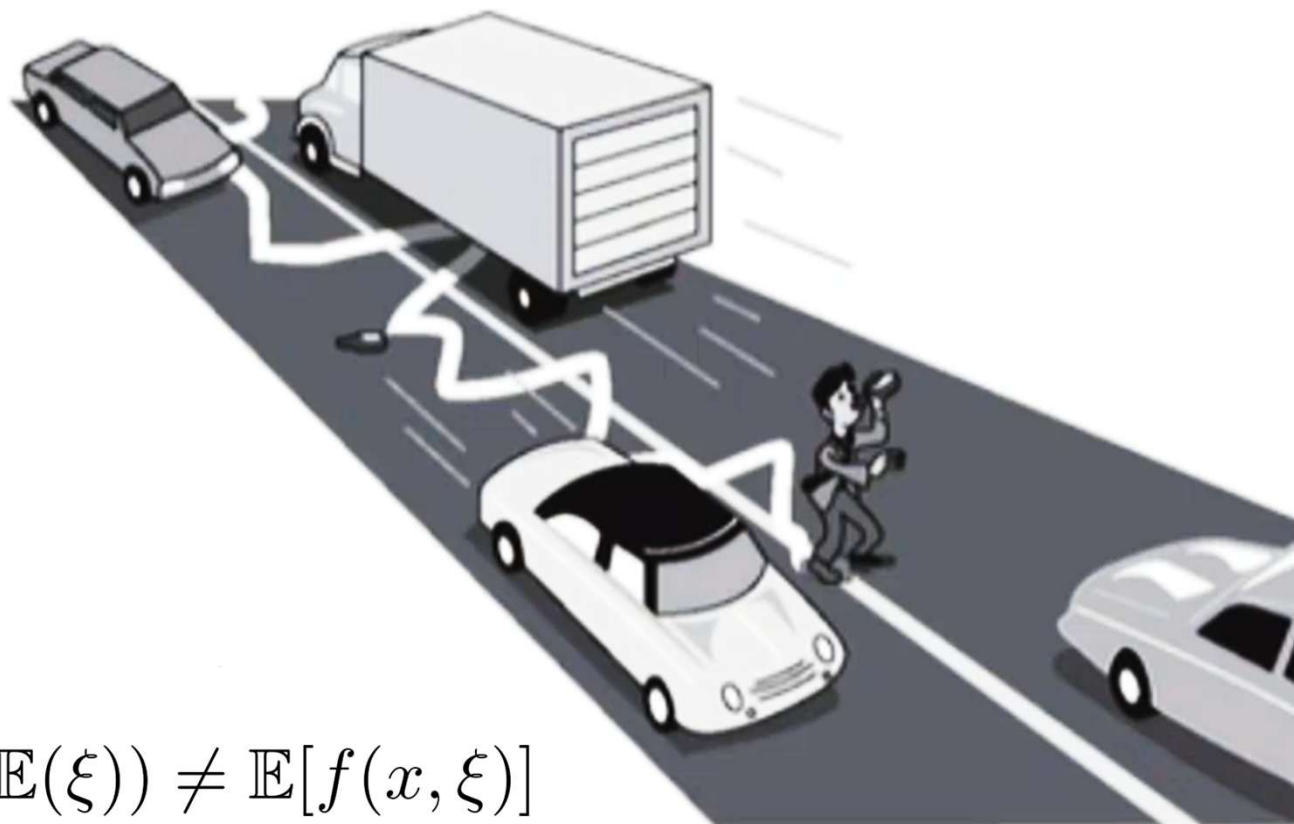
$$x_1, x_2 \geq 0$$



$$x_1^* = 36, x_2^* = 18$$

$$\text{Cost}^* = 126$$

- The **Expected-Value** (EV, or certainty-equivalent) problem

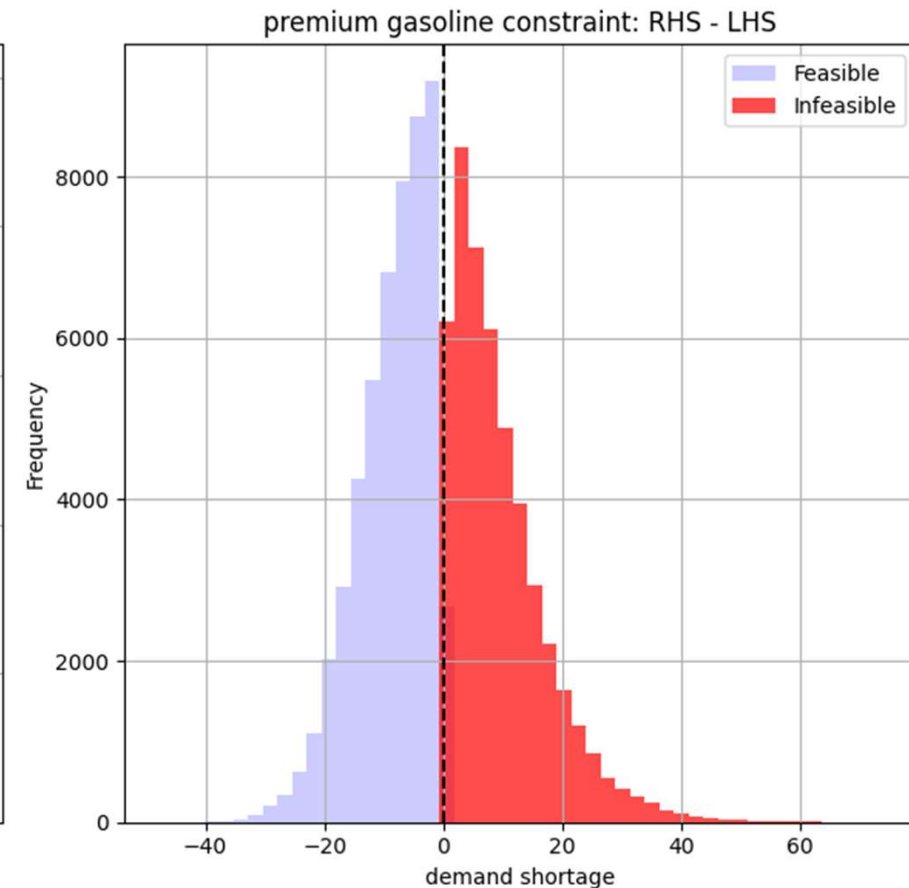
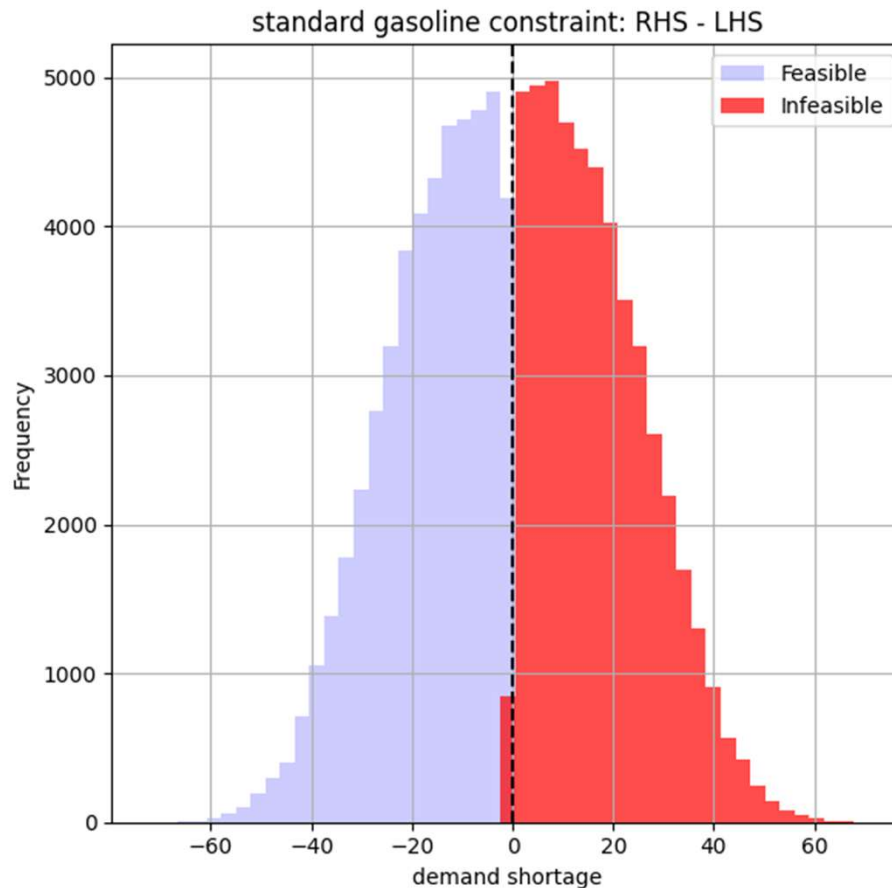


$$f(x, \mathbb{E}(\xi)) \neq \mathbb{E}[f(x, \xi)]$$

- The result (state) of the drunk at his **average** position is **alive**
- But, the **average result** (state) of the drunk is **dead**...



- Overall infeasibility ratio: 73.66%





- A general optimization problem with uncertainty

$$\begin{aligned} \min_x \quad & f_0(x, \xi) \\ \text{s.t.} \quad & f_i(x, \xi) \leq 0, \quad i = 1, \dots, m \end{aligned}$$

- Decision x must be made with **uncertain** parameter ξ
- Applications to many technical and real-world problems
 - Energy, financial, transportation and logistics, healthcare, supply chain, production and manufacturing...
 - Supervised machine learning, reinforcement learning...

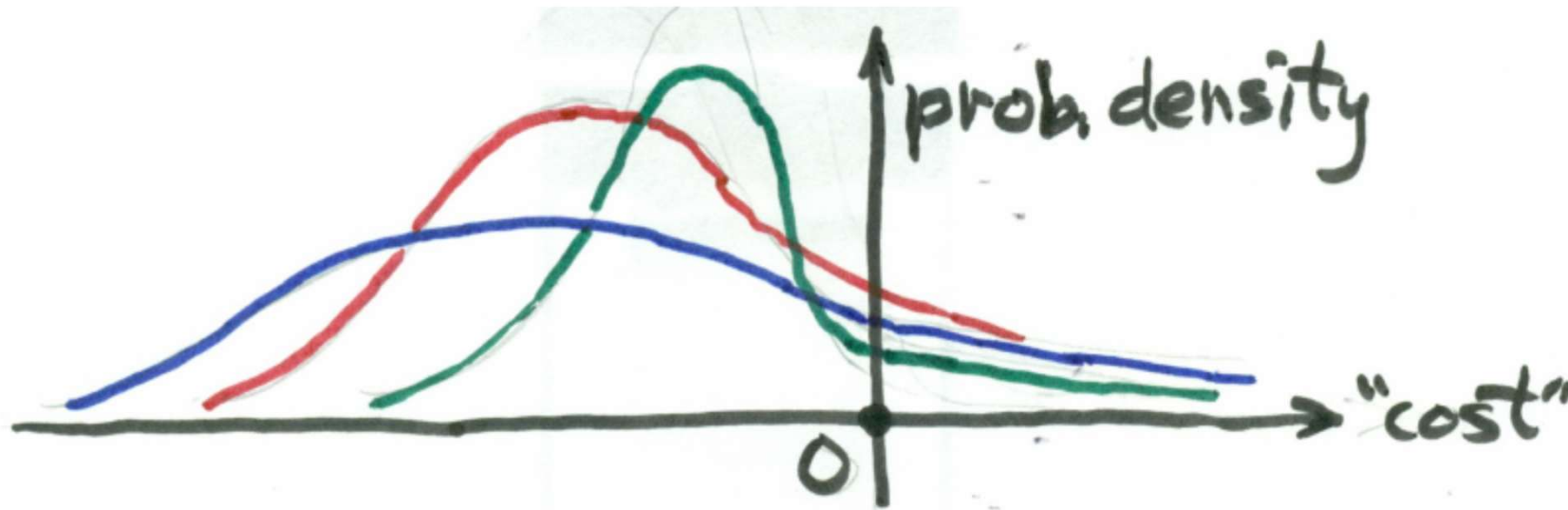


- Stochastic Programming (SP/SO) provides a systematic approach for handling uncertainty, modeling r.v. $\xi \in \Xi$ with **probability measures**.
- Sounds limiting, but
 - An era of “big data”, empirical distribution
 - No need to know the distribution for all uncertain data (“important” ones only)
 - Discretization (“scenarios”) are natural: “low” “medium” “high” demand; weather is “wet” or “dry”
 - Modeling is a subjective matter
- After solving the SP, **always assess solution quality**



Key issues in problem formulation

- The distribution of $f_i(x, \xi)$ can only be shaped by the choice of x
- How then can **constraints** and **minimization** be understood?





Pierre's planting plan



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- Pierre can plant in his land corn, wheat, or sugar beets
- Simply assume the coming season weather will be either good (with probability p) or bad ($1 - p$)
- How should Pierre allocate his land to maximize his profit?

 Corn?	 Beets?
 Wheat?	

	Wheat	Corn	Beets
$\phi_{\text{good}}(\text{K€})$	100	70	80
$\phi_{\text{bad}}(\text{K€})$	-10	40	35

$$\begin{aligned} \max_{x_w, x_c, x_b} \quad & \phi(x_w, x_c, x_b, \xi) \\ \text{s.t.} \quad & x_w + x_c + x_b \leq 1 \\ & x_w, x_c, x_b \geq 0 \end{aligned}$$



- **Wait-and-see** solution (know **perfectly** the coming weather):

	Wheat	Corn	Beets
$\phi_{\text{good}}(\text{K€})$	100	70	80
$\phi_{\text{bad}}(\text{K€})$	−10	40	35

- All wheat if good weather; all corn if bad; its **expected profit**:

$$\phi^{\text{PI}} = \mathbb{E}_{\xi} \left[\max_{\mathbf{x}(\xi)} \phi(\mathbf{x}(\xi)) \right] = 100p + 40(1 - p) = 70 \text{ if } p = 0.5$$

- **But** he must make the plan **here-and-now** before the uncertain weather takes place (with p)...
- Reasonable to plant half wheat and half corn?



	Wheat	Corn	Beets
$\phi_{\text{good}}(\text{K€})$	100	70	80
$\phi_{\text{bad}}(\text{K€})$	-10	40	35

$$\begin{aligned}\mathbb{E}_{\xi} [\phi(x_w = 0.5, x_c = 0.5, \xi)] &= (50 + 35)p + (-5 + 20)(1 - p) \\ &= 50 \text{ if } p = 0.5\end{aligned}$$

- Can we do better?
- Planting **all beets** gives the expected profit = 57.5!

$$\begin{aligned}\max_{x_w, x_c, x_b} \quad & \mathbb{E}[\phi(x_w, x_c, x_b, \xi)] && = (-10 + 110p)x_w \\ & && + (40 + 30p)x_c \\ \text{s.t.} \quad & x_w + x_c + x_b \leq 1 && + (35 + 45p)x_b \\ & x_w, x_c, x_b \geq 0\end{aligned}$$

- A well-defined **SP model with the expectation remedy!**



- Averaging scenario solutions is not a good idea!
- The SP solution = the EV solution for this specific problem, because
 - linear objective function
 - no uncertainty in the constraints
- Life would be easier for Pierre if, before deciding which crop to plant, he could **wait and see** what the weather is...
- The maximum price it is worth, **expected value of perfect information**:

$$\mathbf{EVPI} = \phi^{\mathbf{PI}} - \phi^{\mathbf{SP}} = 70 - 57.5 = 12.5$$



How about feasibility?



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- The stochastic gasoline production problem

$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$(2 + \eta_1)x_1 + 6x_2 \geq 180 + \theta_1$$

$$3x_1 + (3.4 - \eta_2)x_2 \geq 162 + \theta_2$$

$$x_1, x_2 \geq 0$$

- Here-and-now decision variables (x_1, x_2)



A “safe” production plan



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- Look for values of (x_1, x_2) that are **feasible for all possible realizations** of the productivities and demands → **fat solution** (worst-case, risk-aversion)

$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$(2 + \eta_1)x_1 + 6x_2 \geq 180 + \theta_1 \quad \forall \eta_1, \theta_1$$

$$3x_1 + (3.4 - \eta_2)x_2 \geq 162 + \theta_2 \quad \forall \eta_2, \theta_2$$

$$x_1, x_2 \geq 0$$

- A well-defined problem (robust optimization)



A “safe” production plan



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- By taking the **extreme values** of the uncertain parameters, it becomes tractable

$$\min \quad 2x_1 + 3x_2$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$(2 + \eta_1^{\min})x_1 + 6x_2 \geq 180 + \theta_1^{\max}$$

$$3x_1 + (3.4 - \eta_2^{\max})x_2 \geq 162 + \theta_2^{\max}$$

$$x_1, x_2 \geq 0$$

- Solution $(x_1^{\text{wc}}, x_2^{\text{wc}}) = (48.491, 25.453)$ with $\text{Cost}^{\text{wc}} = 173.342$
- Worst-case approach: **expensive and probably too conservative**



- Asking for **feasibility for any future realization** of uncertainty might be too restrictive.
- Depending on the data, extreme rare events may exist, and they can make the (a.s.) feasible set **empty**.
- **Alternatives:**
 - Define a model for which extreme events do not constrain “too much”
models with recourse (decisions)
 - Consider points that are feasible with only some probability level
models with chance-constraints



Instead of asking feasibility for all events, we can allow shortages, charging a penalty for constraint violation

- TotalEnergies agrees with the distribution company in paying a **penalty** if there is a **shortage** in production
- Penalty = price paid if it has to buy the missing gasoline directly in the market, e.g., $[c_1^p, c_2^p]^\top = [7, 12]^\top$
- Amount of penalty can be determined **AFTER** observing the realization of uncertainty
- **Recourse decisions:** amounts of unsatisfied (standard and premium) gasolines for penalty

$y_1(\xi), y_2(\xi)$ for each event $\xi = [\eta_1, \eta_2, \theta_1, \theta_2]$



$$\left\{ \begin{array}{l} x_1 + x_2 \leq 100 \\ \alpha(\xi)x_1 + 6x_2 + y_1(\xi) \geq h_1(\xi) \\ 3x_1 + \beta(\xi)x_2 + y_2(\xi) \geq h_2(\xi) \\ x_1, x_2 \geq 0, y_1(\xi), y_2(\xi) \geq 0 \end{array} \right.$$

- For any point (x_1, x_2) that satisfies the deterministic capacity constraint, there are **wait-and-see** variables that make the point feasible for the demand constraints.
- Now, asking for a.s. feasible points is not a restrictive condition



- Now consider a **fixed event** $\tilde{\xi}$, the deterministic LP is:

$$\begin{aligned} \min \quad & 2x_1 + 3x_2 + 7y_1(\tilde{\xi}) + 12y_2(\tilde{\xi}) \\ \text{s.t.} \quad & x_1 + x_2 \leq 100 \\ & \alpha(\xi)x_1 + 6x_2 + y_1(\tilde{\xi}) \geq h_1(\xi) \\ & 3x_1 + \beta(\xi)x_2 + y_2(\tilde{\xi}) \geq h_2(\xi) \\ & x_1, x_2 \geq 0, y_1(\tilde{\xi}), y_2(\tilde{\xi}) \geq 0 \end{aligned}$$

- We need to choose one feasible point (x_1, x_2) **here-and-now**, that is **satisfactory** according to some criterion.



- **Question:** having chosen the values (x_1, x_2) , what can I expect in future when uncertain event take place?
- For **one** future event $\tilde{\xi}$, given the decision (x_1, x_2) ,

$$C(x_1, x_2, \tilde{\xi}) = \begin{cases} \min & 2x_1 + 3x_2 + 7y_1(\tilde{\xi}) + 12y_2(\tilde{\xi}) \\ \text{s.t.} & x_1 + x_2 \leq 100 \\ & \alpha(\xi)x_1 + 6x_2 + y_1(\tilde{\xi}) \geq h_1(\xi) \\ & 3x_1 + \beta(\xi)x_2 + y_2(\tilde{\xi}) \geq h_2(\xi) \\ & x_1, x_2 \geq 0, y_1(\tilde{\xi}), y_2(\tilde{\xi}) \geq 0 \end{cases}$$

is the optimal cost, and

it is a **random variable** if we let $\tilde{\xi}$ vary



The **expectation** remedy:

$$\min_{x_1, x_2} \mathbb{E}[C(x_1, x_2, \xi)] =$$
$$\left\{ \begin{array}{ll} \min & 2x_1 + 3x_2 + \mathbb{E}[7y_1(\xi) + 12y_2(\xi)] \\ \text{s.t.} & x_1 + x_2 \leq 100 \\ & \alpha(\xi)x_1 + 6x_2 + y_1(\xi) \geq h_1(\xi) \\ & 3x_1 + \beta(\xi)x_2 + y_2(\xi) \geq h_2(\xi) \\ & x_1, x_2 \geq 0, y_1(\xi), y_2(\xi) \geq 0 \end{array} \right\} a.s. [\xi]$$

- A **valid** SP model
- For **general** distribution of ξ
 - Computing the expectation can be very expensive!
 - There may be **infinitely many** constraints!



- In very important special cases, ξ has a **finite** support:

$$\xi \in \{\xi_1, \xi_2, \dots, \xi_N\} \text{ where } \xi_i \text{ has probability } p_i$$

- The model with recourse is

$$\min \quad 2x_1 + 3x_2 + \sum_{i=1}^N p_i [7y_1(\xi_i) + 12y_2(\xi_i)]$$

$$\text{s.t.} \quad x_1 + x_2 \leq 100$$

$$x_1, x_2 \geq 0$$

$$\left. \begin{array}{l} \alpha(\xi_i)x_1 + 6x_2 + y_1(\xi_i) \geq h_1(\xi_i) \\ 3x_1 + \beta(\xi_i)x_2 + y_2(\xi_i) \geq h_2(\xi_i) \\ y_1(\xi_i), y_2(\xi_i) \geq 0 \end{array} \right\} i = 1, \dots, N$$



- We sample $N = 100$ scenarios from the uncertain parameter distributions (a crude approximation)
- How many recourse decision variables and constraints?
- An optimal solution for this LP

$$x^{100} = (39.384, 21.752)$$

- Its production cost = 144.023 and its **expected penalty cost** = 5.322

$$\mathbb{P} \left[\begin{array}{l} \alpha(\xi)x_1^{100} + 6x_2^{100} \geq h_1(\xi) \\ 3x_1^{100} + \beta(\xi)x_2^{100} \geq h_2(\xi) \end{array} \right] \approx 0.924$$



A brief comparison



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- Assessing the quality of the solution

Model	x_1	x_2	Production cost	Feasibility
Deterministic (EV)	36	18	126	26.4%
Worst-case (RO)	48.491	25.453	173.374	~ 100%
Recourse model	39.384	21.752	144.023	~ 92.4%

- The recourse model: not too expensive decisions, with reasonable probability of being feasible



- For a minimization problem with uncertain parameters

$$\min_{x \in \mathcal{X}(\xi)} C(x, \xi)$$

Perfect information model

$$x_{PI}^*(\xi) = \operatorname{argmin}_{x(\xi) \in \mathcal{X}(\xi)} C(x(\xi), \xi) \quad C^{PI} = \mathbb{E}_{\xi} [C(x_{PI}^*(\xi), \xi)]$$

Expected value model

$$x_{EV}^* = \operatorname{argmin}_{x \in \mathcal{X}(\mathbb{E}[\xi])} C(x, \mathbb{E}[\xi]) \quad C^{EV} = \mathbb{E}_{\xi} [C(x_{EV}^*, \xi)]$$

Stochastic program model

$$x_{SP}^* = \operatorname{argmin}_{x \in \mathcal{X}(\xi)} \mathbb{E}_{\xi} [C(x, \xi)] \quad C^{SP} = \mathbb{E}_{\xi} [C(x_{SP}^*, \xi)]$$



- Expected value of perfect information:

$$\text{EVPI} = \mathbf{C}^{\text{SP}} - \mathbf{C}^{\text{PI}} = \mathbb{E}_{\xi} [C(x_{\text{SP}}^*, \xi)] - \mathbb{E}_{\xi} [C(x_{\text{PI}}^*(\xi), \xi)] \geq 0$$

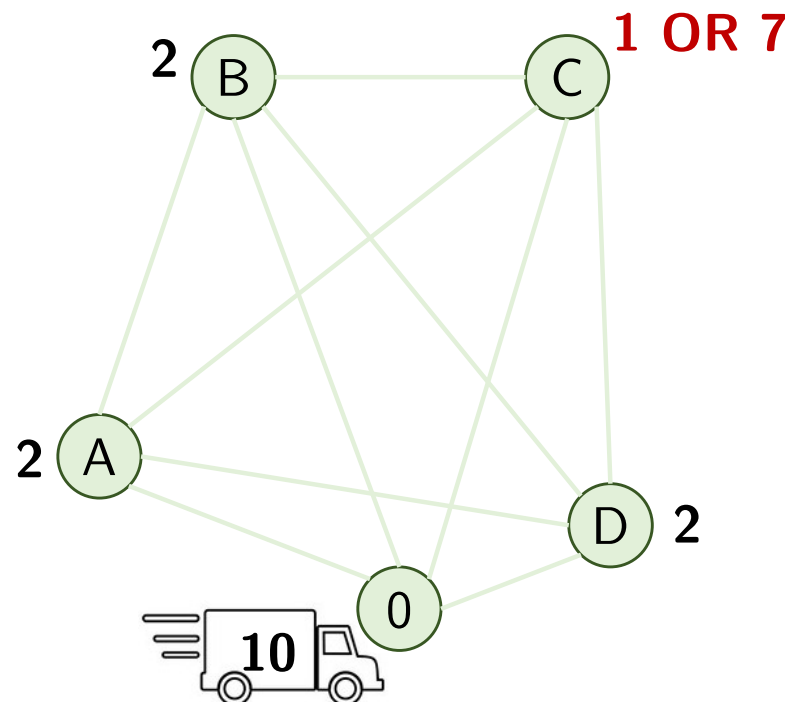
- Value of SP solution:

$$\text{VSS} = \mathbf{C}^{\text{EV}} - \mathbf{C}^{\text{SP}} = \mathbb{E}_{\xi} [C(x_{\text{EV}}^*, \xi)] - \mathbb{E}_{\xi} [C(x_{\text{SP}}^*, \xi)] \geq 0$$



A routing example

- Serve clients at A, B, C, and D from depot (0) with vehicle of capacity 10
- Demand = 2 at A, B, D
- Demand = 1 OR 7 at C



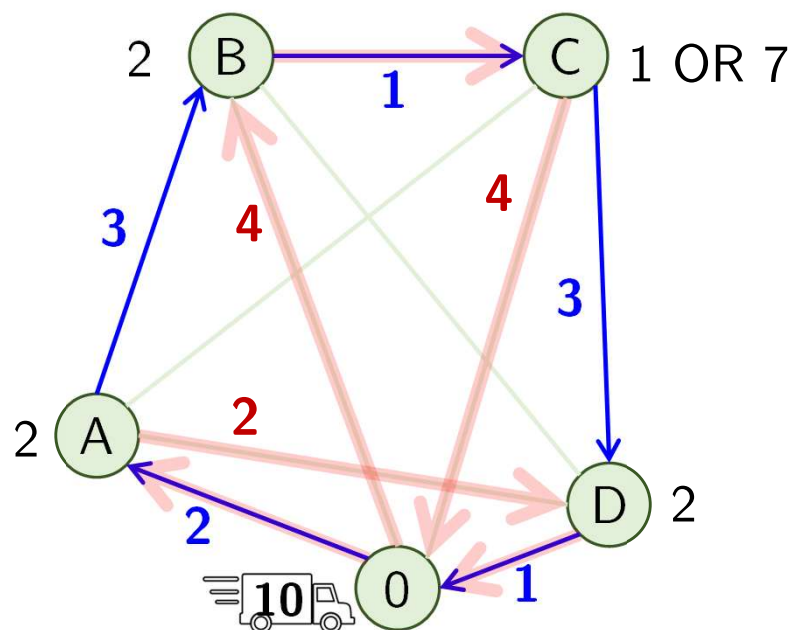
	0	A	B	C	D
0	—	2	4	4	1
A	2	—	3	4	2
B	4	3	—	1	3
C	4	4	1	—	3
D	1	2	3	3	—



A routing example



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	0	A	B	C	D
0	—	2	4	4	1
A	2	—	3	4	2
B	4	3	—	1	3
C	4	4	1	—	3
D	1	2	3	3	—

- Wait-and-See solution:

- $\xi=1$: $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow 0$ of length 10
- $\xi=7$: $0 \rightarrow A \rightarrow D \rightarrow 0 \rightarrow B \rightarrow C \rightarrow 0$ of length 14

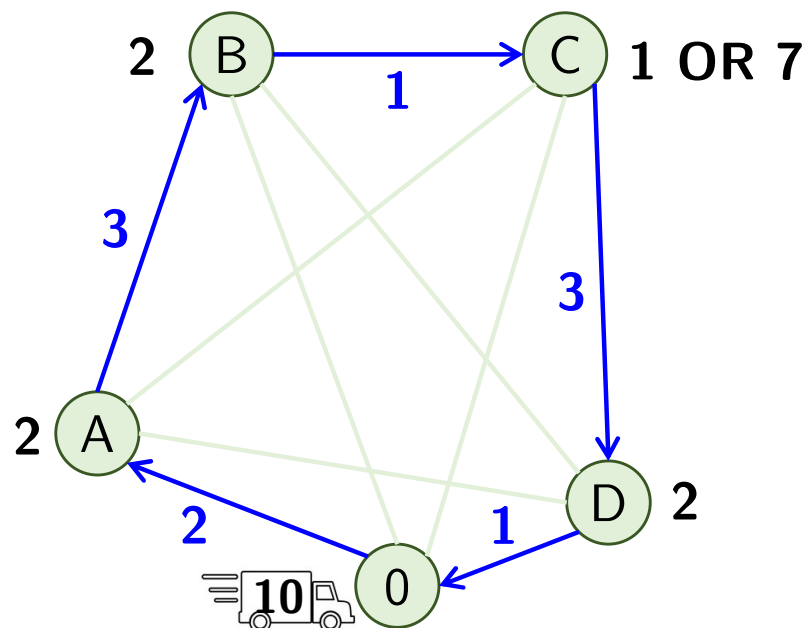
$$C^{PI} = \mathbb{E}_{\xi} [C(x_{PI}^*(\xi), \xi)] = \frac{10 + 14}{2} = 12$$



A routing example



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	0	A	B	C	D
0	—	2	4	4	1
A	2	—	3	4	2
B	4	3	—	1	3
C	4	4	1	—	3
D	1	2	3	3	—

- Expected value (EV) solution: $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow 0$ (or inverse direction) of length (cost) 10
- Expected cost of EV solution depends on direction (14 or 13.5)

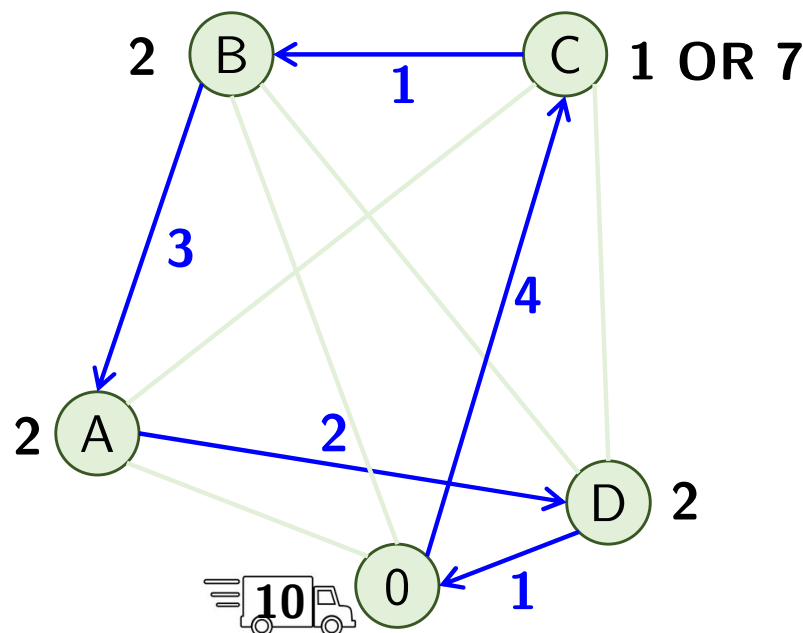
$$C^{\text{EV}} = \mathbb{E}_{\xi} [C(x_{\text{EV}}^*, \xi)] = 14 \text{ or } 13.5$$



A routing example



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	0	A	B	C	D
0	—	2	4	4	1
A	2	—	3	4	2
B	4	3	—	1	3
C	4	4	1	—	3
D	1	2	3	3	—

- Stochastic program (SP) solution: $0 \rightarrow C \rightarrow B \rightarrow A \rightarrow D \rightarrow 0$
 - $\xi=1$: a length of 11
 - $\xi=7$: a length of 14

$$C^{SP} = \mathbb{E}_{\xi} [C(x_{SP}^*, \xi)] = \frac{11 + 14}{2} = 12.5$$



- Expected value of perfect information:

$$EVPI = C^{SP} - C^{PI} = 12.5 - 12 = 0.5$$

- Value of SP solution:

$$VSS = C^{EV} - C^{SP} = 14 \text{ or } 13.5 - 12.5 = 1.5 \text{ or } 1$$

- Larger VSS than EVPI
- EV deterministic solution $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow 0$ is
 - wrongly (wishful 10, but turn out to be 14/13.5)
 - optimistic ($10 < 12.5$)



- **Uncertainty** is everywhere, posing significant challenges in optimization
- EV model is simple, but not good enough
- Worst case (RO) model probably too conservative
- SP approach with **probabilistic modeling** of uncertainty
 - expectation remedy
 - recourse models
- Value of information: VSS and EVPI **both nonnegative**, can have **larger VSS**

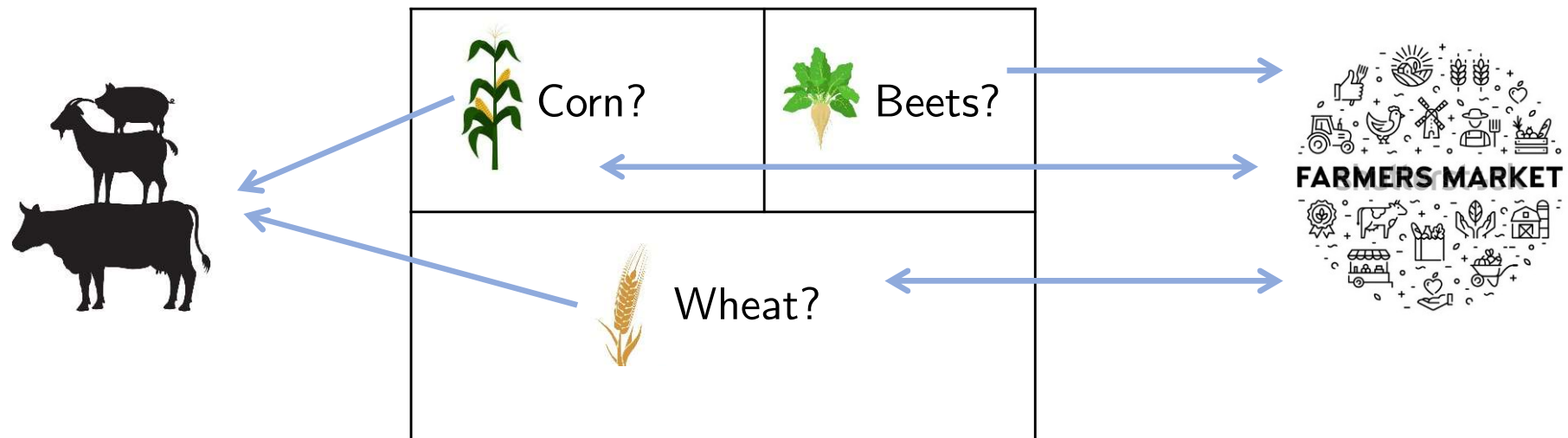


Exercise: Pierre's planting plan



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- Pierre has **500 hectare** of land
- He grows **wheat**, **corn** and **sugar beets**, also has some **cattle** to feed requiring a certain amount of wheat and corn
- Sell crops when there is an excess and buy them (with higher prices) when there is a shortage
- Pierre's winter problem: how much land to devote to each crop?



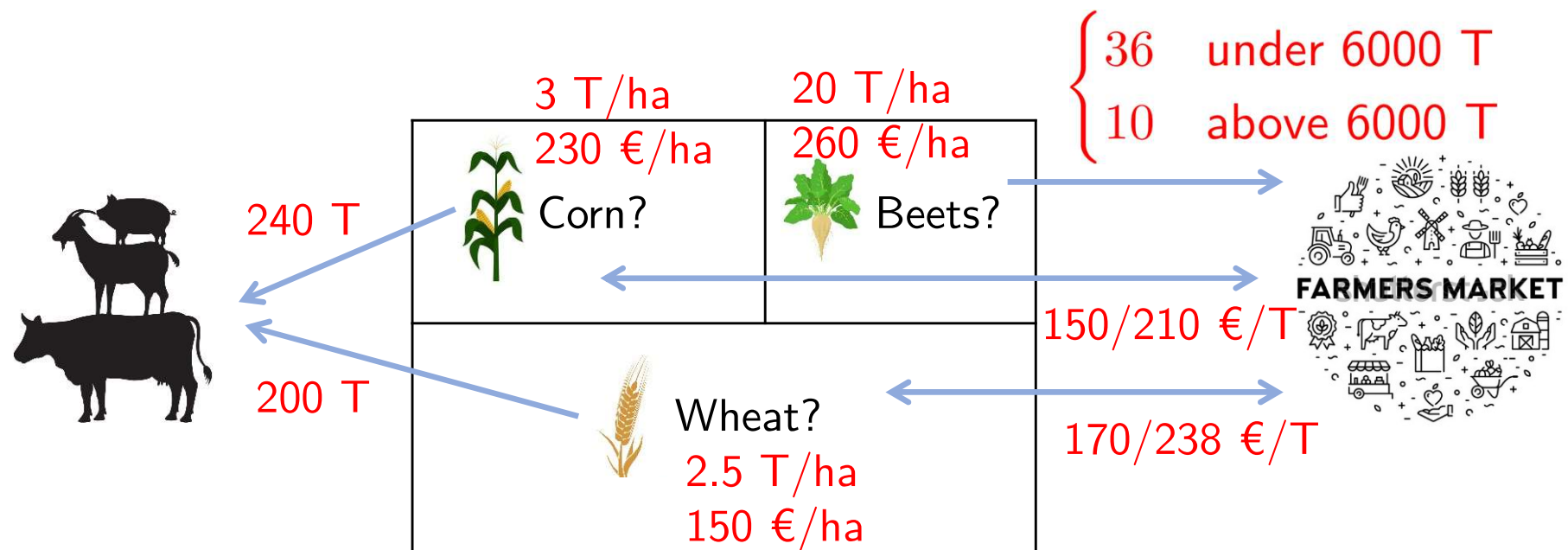


Exercise: Pierre's planting plan



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- Pierre knows the **mean yield**, the **planting cost** of each crop (per ha), and **the needs** of corn and wheat.
- The selling prices on the market for the wheat and corn are known
- He can **sell up to 6000 tons** of beets at a favorable price, and the rest at a discounted price





- **Yield uncertainty**: crop yields have been very different from year to year, due to weather conditions, pests, seed quality, ...

	Wheat	Corn	Sugar Beets
Yield (T/ha)	2.5	3	20
Planting cost (€/ha)	150	230	260
Sell price (€/T)	170	150	36 under 6000 T 10 above 6000T
Purchase price (€/T)	238	210	-
Min demand (T)	200	240	-
Total available land: 500 ha			

$\pm 20\%$

- Simplifying assumptions: years are **good (+20%)**, **fair (100%)**, or **terrible (-20%)** for all crops with even probability
- Pierre wants to **maximize the profit** (selling earnings — planting and purchase expenses)



- 1) Formulate the EV deterministic problem
- 2) Scenario analysis
- 3) Formulate the SP model with the expectation remedy
- 4) Calculate the EVPI and VSS



- Why and under which condition that the Expected Value model solution is wrongly optimistic?



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